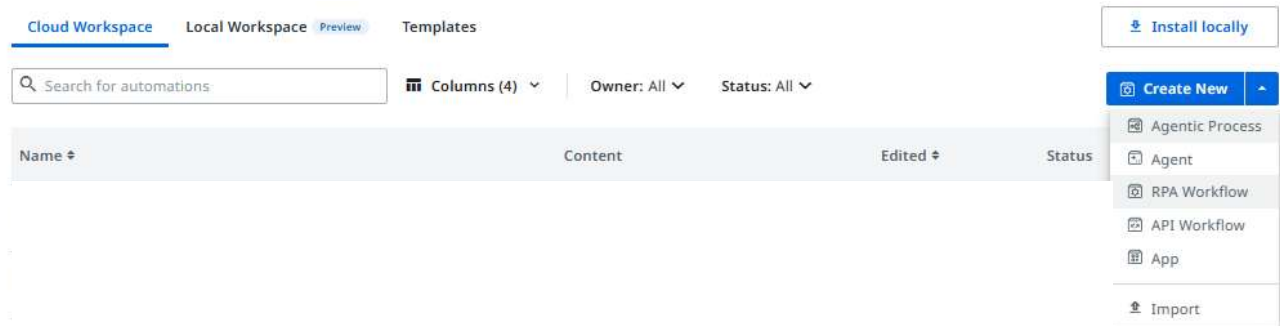


Week 3 Automating Document Data extraction

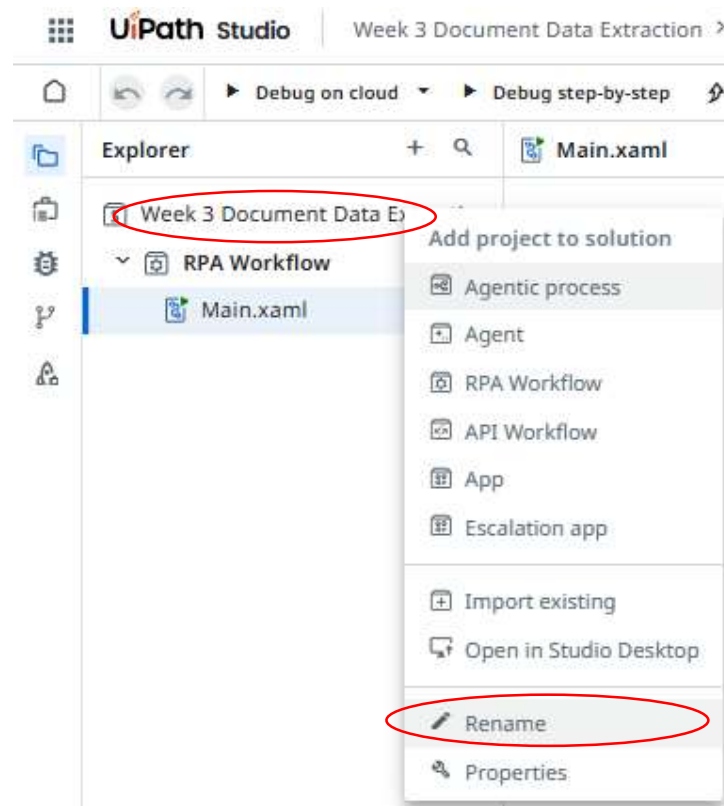
Pre-Work

- a. Copy input data folder and output Google spreadsheet into your own Google drive

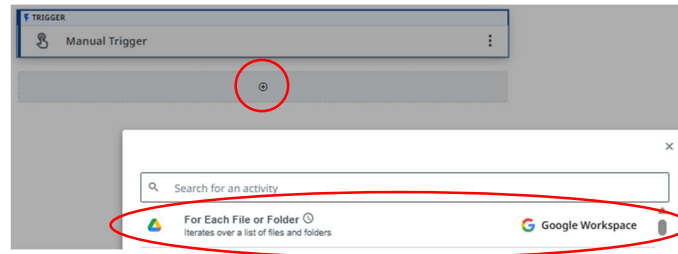
Click Create New → RPA Workflow



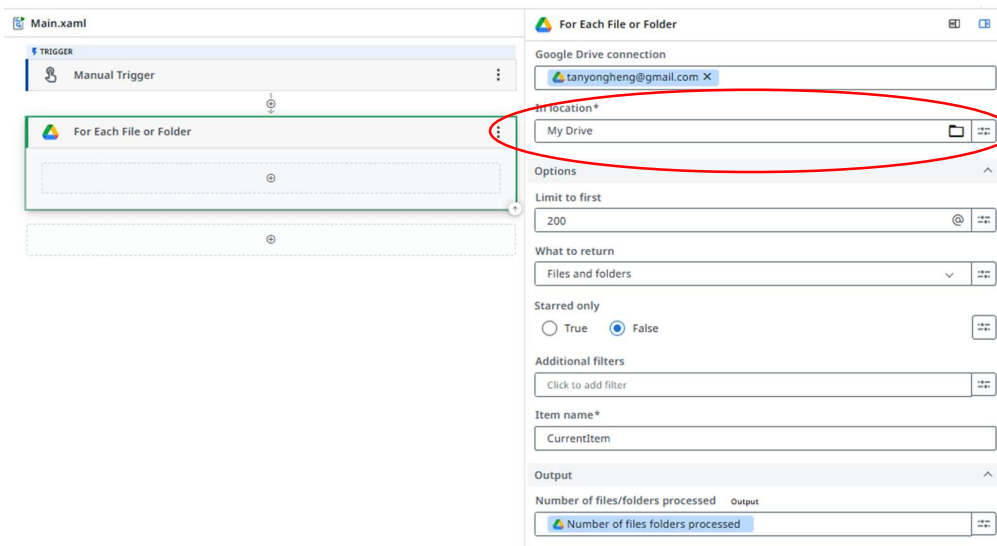
1. Right click “Solution” and Rename it to “Document Data Extraction”
-



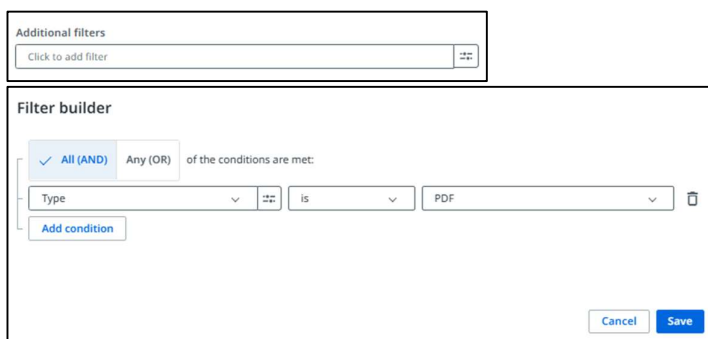
2. Add a For Each file or folder activity (Google Workspace).



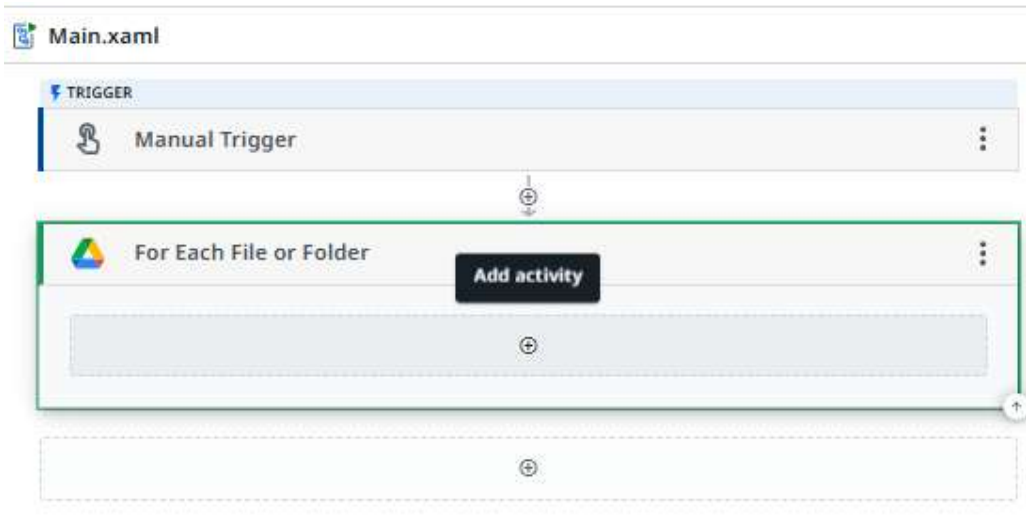
3. For the "In location" field, traverse to the folder containing the input files and select it.



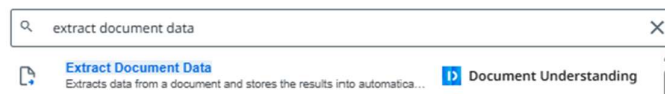
4. Under “Additional Filter”, click on field that says “Click to add filter” and Choose Type is PDF.



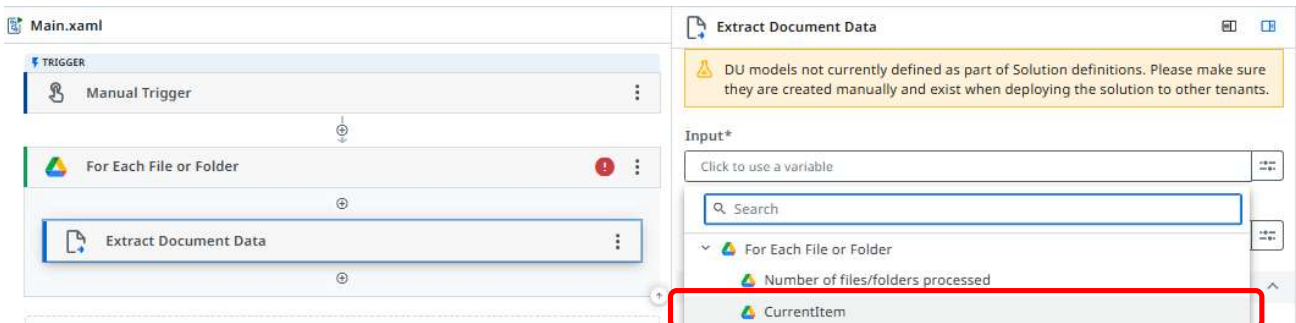
5. Once the filter condition is set, click on the + icon to add another activity.



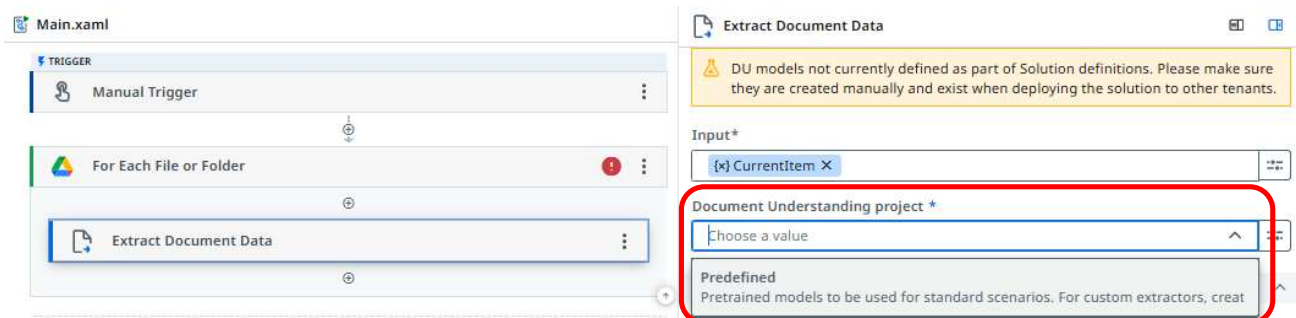
Search for “Extract Document Data” and select the activity. This is part of the UiPath Document Understanding framework and is used to extract data from a file, storing the results in an automatically generated output variable called “Document Data.”



In the **Input** field, choose **CurrentItem**. This refers to the current input file to be processed.



In the **Document Understanding project** field, choose **Predefined**



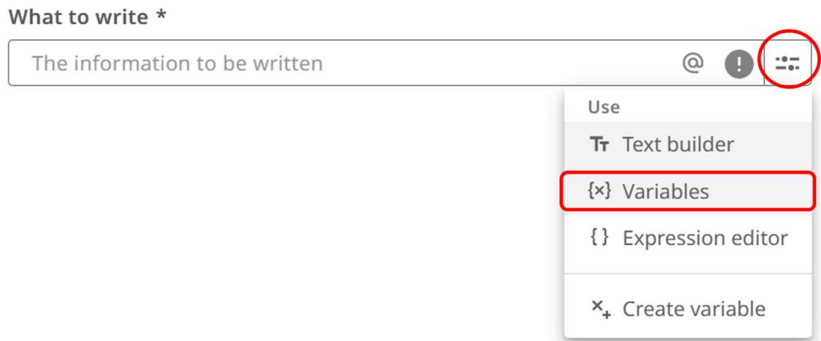
For "Extractor" select "Invoices" from the drop-down. Observer that there are various extractors which are essentially out of the box ML models designed to extract data from specific documents.

The screenshot shows a workflow editor with a canvas on the left and a configuration panel on the right. The canvas has three steps: 'Manual Trigger', 'For Each File or Folder', and 'Extract Document Data'. The 'Extract Document Data' step is selected. The configuration panel on the right is titled 'Extract Document Data' and contains a warning message: 'DU models not currently defined as part of Solution definitions. Please make sure they are created manually and exist when deploying the solution to other tenants.' Below the warning, there are four fields: 'Input*' with a dropdown showing '{x} CurrentItem X', 'Document Understanding project *' with a dropdown showing 'Predefined', 'Extractor *' with a dropdown showing 'Invoices' (highlighted with a red box), and 'What to write*' with a dropdown showing 'Data Vendor name Value X'.

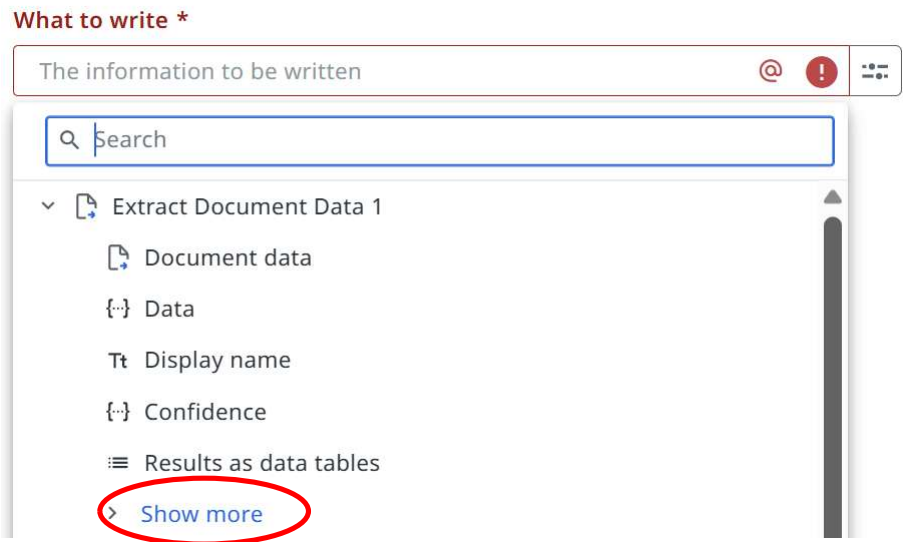
7. Add a Write Cell activity (Google Workspace).
 - a. SpreadSheet: DataExtraction_Output
 - b. Sheet: Individual
 - c. Cell: A1

The screenshot shows the 'Write Cell' activity configuration in Google Workspace. At the top, there is a search bar with the text 'write cell' and a magnifying glass icon. Below the search bar, there is a card for 'Write Cell' with the description 'Writes information in the selected spreadsheet cell' and the Google Workspace logo. Below the card, there is a configuration panel for the 'Write Cell' activity. The panel has a title 'Write Cell' and a subtitle 'Google Sheets connection'. It contains several fields: 'Spreadsheet*' with a dropdown showing 'DataExtraction_Output', 'Sheet or named range *' with a dropdown showing 'Individual', 'Cell' with a dropdown showing 'A1', and 'What to write*' with a dropdown showing 'Data Vendor name Value X'.

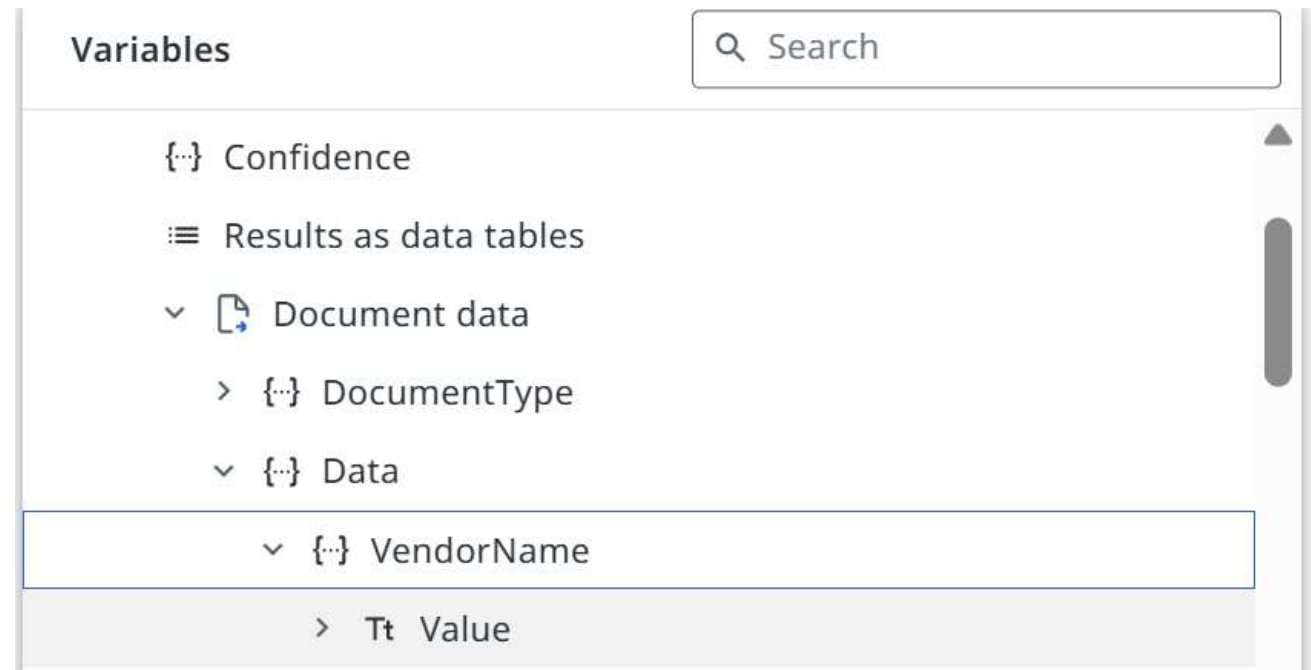
In what to write field, click “see more” → “variables”



Choose “Show more”



Select Document Data → Data → VendorName → Value



Rename “Write Cell” to “Write Cell – Vendor Name”



8. Add a Write Cell activity (Google Workspace)

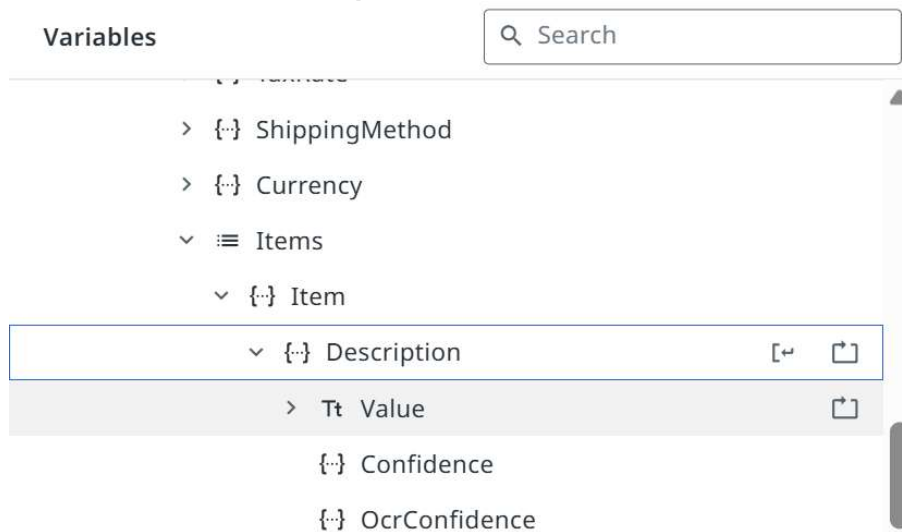
- Cell: B1
- What to Write: Show More → Document Data → Data → InvoiceNumber → Value
- Rename to: Write Cell – Invoice Number

9. Add a Write Cell activity (Google Workspace).

- Cell: C1
- What to Write: Show More → Document Data → Data → InvoiceDate → Value
- Rename to: Write Cell – Invoice Date

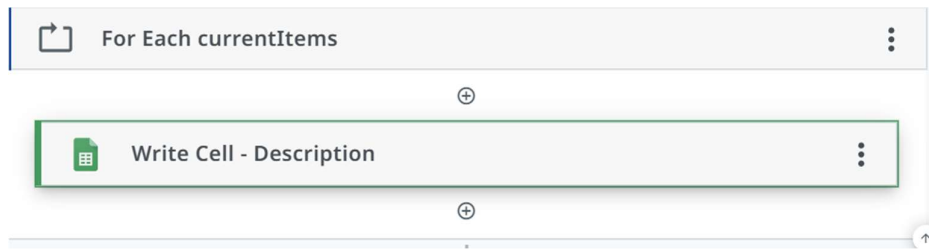
10. Add a Write Cell activity (Google Workspace).

- Cell: (leave blank)
- What to Write: Show More → Document Data → Data → Items → Item → Description → Value
- Rename to: Write Cell – Description

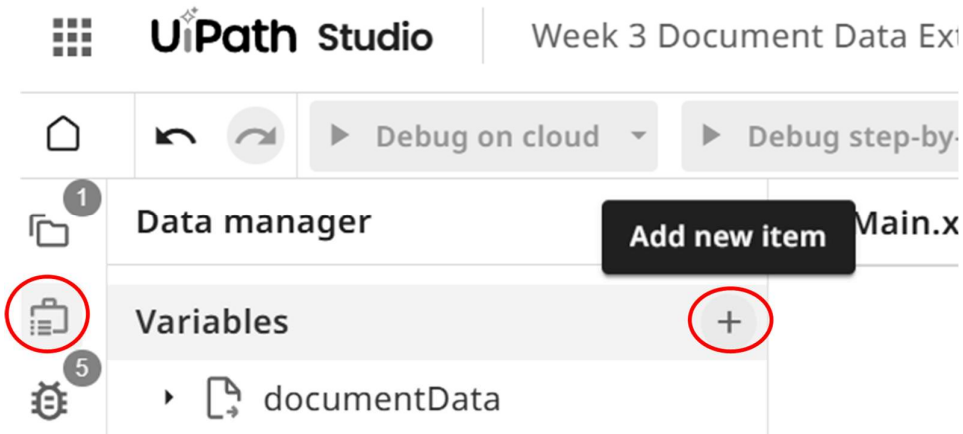


! Notice that the write cell activity is now enclosed in a “For Each” activity, which has been automatically added.

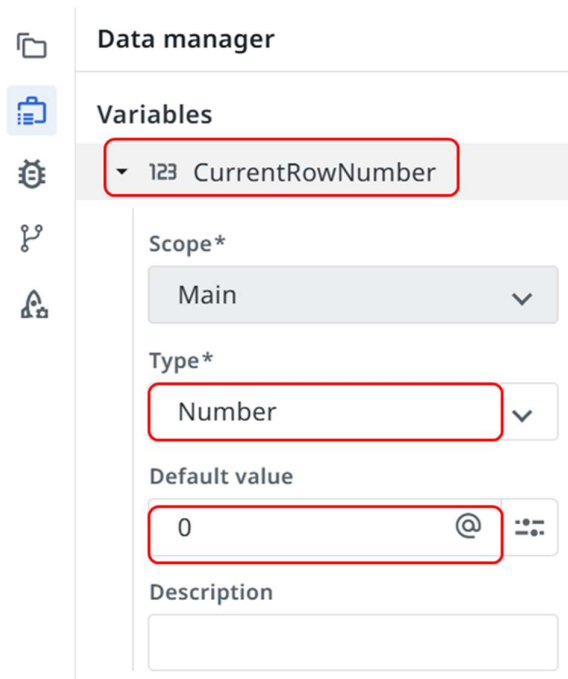
This is because “item description” contains a “list” of items. To iterate over the list, write cell activity has been enclosed inside a For Each activity.



11. Go to “Data Manager”, click “Add new item”



Name our new variable **CurrentRowNumber**, change the type to **Number** and set default value to **0**



12. Add a new activity “Set Variable Value” before “Write Cell – Description”

**Set Variable Value** ⓘ
Assign a value to a variable, enabling its reuse in subsequent activities within the workflow.

 **Control**

To fill in the **To variable** field, Click **See More** and choose **Change Variable**

**Assign Variable Value**

To variable* Output

Click to use a variable

Set value *

Enter text

Use

{*} Change variable

{ } Expression editor

*+ Create variable

Choose **CurrentRowNumber** variable

**Assign Variable Value**

To variable* Output

Click to use a variable

123 CurrentRowNumber

For the **Set value** field, click **See More** and select **Expression editor**

**Assign Variable Value**

To variable* Output

{*} CurrentRowNumber X

Set value *

Enter numeric expression

Use

Tr Numeric editor

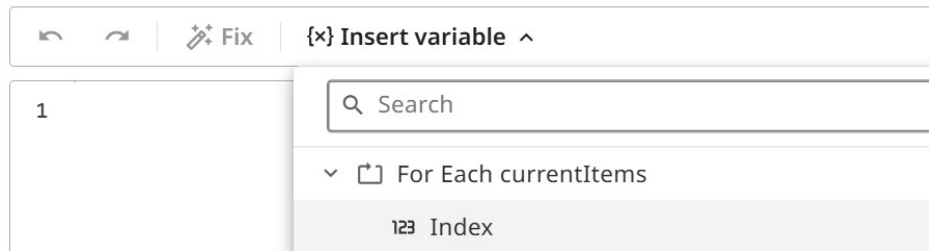
{*} Variables

{ } Expression editor

*+ Create variable

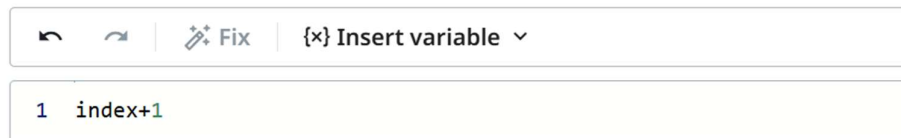
In the **Expression editor**, click **Insert variable** and choose **Index**

VB Expression editor Assign Variable Value > Set value (123 Int32)



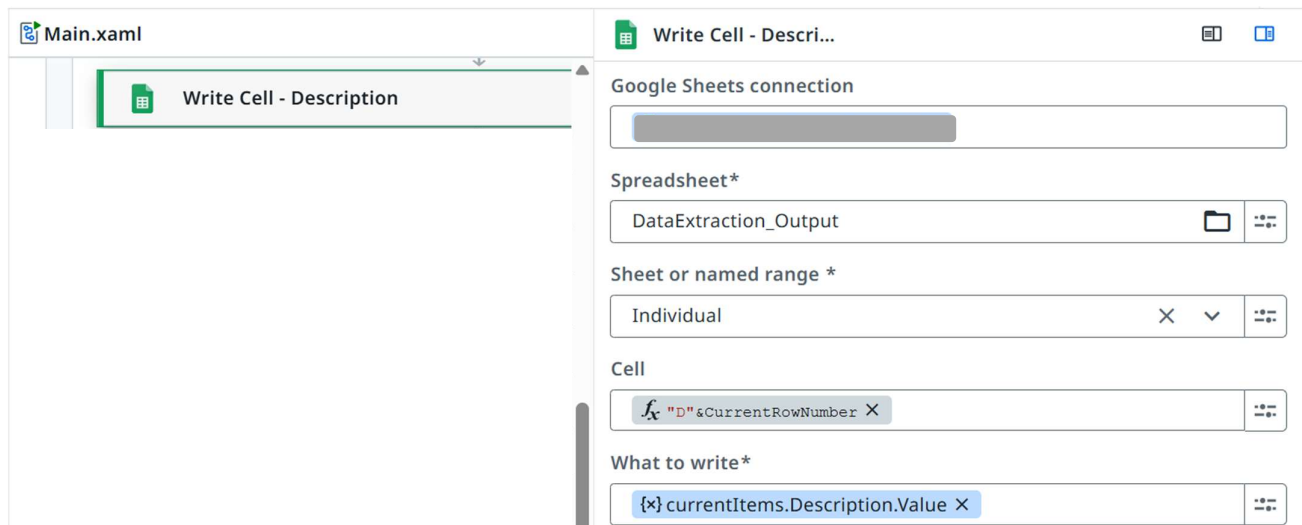
Add one to the index by appending +1 to it, click **Save**

VB Expression editor Assign Variable Value > Set value (123 Int32)



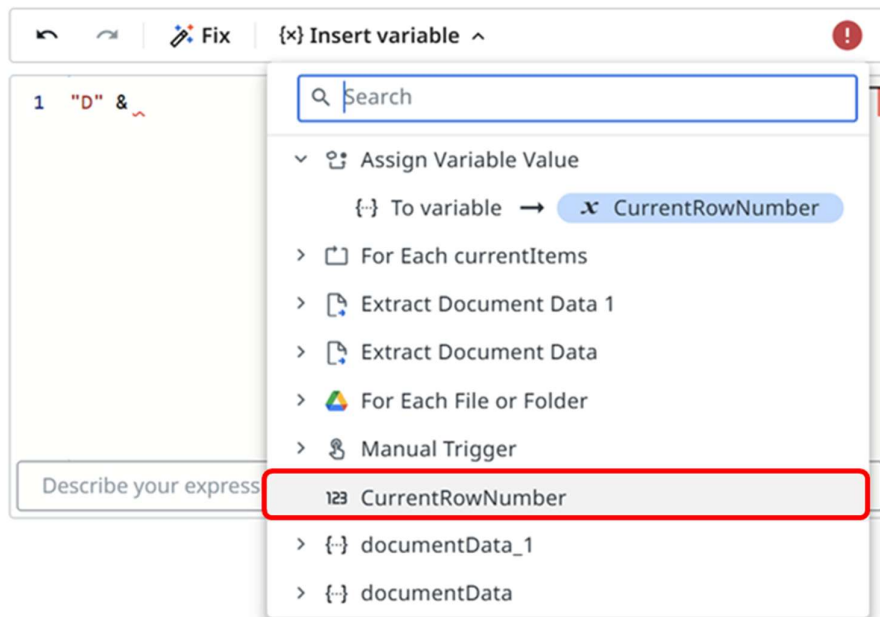
Note: Row count in spreadsheet starts from 1. But in a list, the index starts from 0. To find the current row number's value, we use index+1.

13. Now we focus on attention on “Write Cell – Description”. In the Cell field, click “See More” and select “Expression Editor”



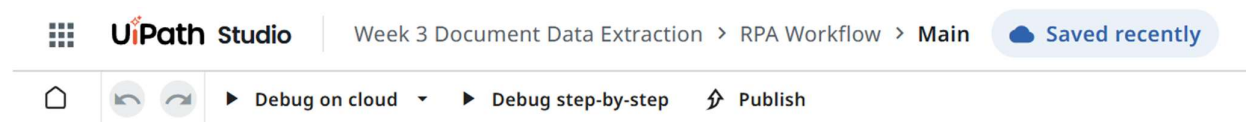
In the Expression editor, type in “D” & <insert variable → **CurrentRowNumber**>

VB Expression editor Write Cell - Description > Cell (Tt String)



14. Repeat step 13 for Quantity, Unit Price and Amount.

15. Run the program by clicking on “Debug on Cloud”



You'll observed that invoice details are being overwritten, with the details of the last invoice being displayed at the end of the workflow execution.

16. Add a “Read Range” activity

**Read Range** ⓘ
Reads and saves the information from a spreadsheet's range for subsequent a...

Fill in the fields:

- Spreadsheet: As per your setup
- Range: ExtractedValue
- Has Header: False

Main.xml

 Read Range

Google Sheets connection

Spreadsheet *

Use spreadsheet template

DataExtraction_Output

Range *

Individual

Has headers

☐ True ☒ False

Input

What to read

Values

Output

Read range

Output

{x} readRange X

Range information

Output

 Range information

17. Add a “Write Range” activity

 Write Range

Write ...

Google Sheets connection
tanyongheng@gmail.com_1 X

Spreadsheet * [Use spreadsheet template](#)
DataExtraction_Output

Range *
Overall

Include headers
☐ True ☒ False

How to write
Append

What to write*
{x} readRange X

18. Add a “Delete Range” activity

 Delete Range

Delete ...

Google Sheets connection
tanyongheng@gmail.com_1 X

Spreadsheet*
DataExtraction_Output

Range *
Individual

Delete behavior
Clear

19. Run the program by clicking on “Debug on cloud”

 **UiPath Studio** | Week 3 Document Data Extraction > RPA Workflow > Main [Saved recently](#)

   ▶ Debug on cloud ▼ ▶ Debug step-by-step ▶ Publish